University of El Oued
Faculty of Exact Sciences – Department of Computer Science
Master 2 – Artificial Intelligence and Data Science
Module: Big Data Analytics

TP04: Containerization and Big Data

Environment Deployment Using Docker

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Academic Year: 2026–2027

1 Objective

The objective of this practical work is to deploy a Python application using Docker and process a large traffic dataset. The project includes building a Docker image, publishing it to Docker Hub, running containers, and creating a custom Docker network.

2 Technologies Used

- Docker and Docker Compose.
- Python 3.12 and Pandas.
- Docker Hub.
- A CSV traffic dataset of approximately 5 GB.

3 Building and Publishing the Docker Image

The Docker image was built and published to Docker Hub using the following commands:

```
docker build -t zekriayoub19/tp04-bigdata:1.1 .
docker push zekriayoub19/tp04-bigdata:1.1
docker pull zekriayoub19/tp04-bigdata:1.1
```

The image was successfully published and downloaded.

4 Data Processing

The application uses Pandas to read the CSV file in chunks of 100,000 rows. This approach reduces the need to load the entire dataset into memory at once.

The application was executed using Docker Compose:

```
docker compose run --rm data-analysis
```

5 Docker Network and Containers

A custom Docker network was created using:

```
docker network create tp04-network
```

Three containers were connected to this network. The `docker ps` command confirmed that all three containers were running.

DNS communication was tested using:

```
docker exec tp04-container1 python -c "import socket; print(socket.
    gethostbyname('tp04-container2'))"
```

The result was `172.20.0.3`, confirming that the second container's name was successfully resolved.

6 Results

Metric	Result
Total number of rows	135,413,619
Number of columns	8
Chunks processed	1,355
Average traffic intensity	432.074
Average sensor occupancy	7.478
Average speed	19.764

The data processing completed successfully. The analysis also identified missing values in some columns of the dataset.

7 Conclusion

This practical work demonstrated the main features of Docker, including image building, publishing images to Docker Hub, chunk-based data processing, volume mounting, and communication between containers.

The application successfully processed more than 135 million rows, demonstrating a practical approach to handling large datasets in a containerized environment.